Department of Information and Communication Engineering (ICE)

Industrial Tanning Report

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Course Title: Industrial Training 01 (Riseup Labs)

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1. Understanding the Software Development Life Cycle (SDLC)

Requirements Gathering: Our industrial training commenced with an insightful session led by Product Manager Rakib Hasan Dipu. He elucidated the nuances of requirement gathering from clients, emphasizing the pivotal role of a product manager in bridging client expectations with development teams. A significant takeaway was understanding the distinction between product and project managers: while project managers focus on the development to delivery phases, product managers encompass a broader scope, including client interactions and post-delivery support. The introduction to Agile methodologies, particularly the concept of "sprints," highlighted how large projects are segmented into manageable units, ensuring streamlined processes and budget adherence.

System Design: Senior Software Developer Alvi Bhaiya conducted an engaging session on system design. He demonstrated how projects transition from conceptualization to development, often originating from research papers or innovative ideas. Sharing his personal achievements, including receiving the Best Paper Award at an IEEE conference in Europe, he showcased the transformation of academic research into commercial applications. This session underscored the importance of a well-structured system design in laying the foundation for successful software development.

Implementation: The implementation phase was brought to life through practical demonstrations by Alvi Bhaiya. He showcased the efficient use of GitHub on a Linux platform, emphasizing best practices in version control and collaborative development. This hands-on experience provided a clear perspective on translating design documents into functional software components.

Testing: Our exploration into the testing phase involved hands-on assignments where we evaluated one of Riseup Labs' live websites. We delved into various testing methodologies, including unit testing, integration testing, system testing, and acceptance testing. This practical exposure highlighted the critical role of quality assurance in delivering robust and reliable software products.

Deployment and Maintenance: In a concise overview, Alvi Bhaiya introduced us to the deployment process, shedding light on the responsibilities of DevOps engineers in launching applications into production environments. He also touched upon maintenance practices, emphasizing the need for continuous monitoring and updates to ensure optimal performance. A more detailed session on this topic is scheduled for our concluding class.

2. Comparison Between Academic Knowledge and Industry Practices

At DIU, our curriculum provides a theoretical understanding of the SDLC phases. However, this industrial attachment at Riseup Labs bridged the gap between theory and practice. Engaging directly with professionals allowed us to witness the real-world application of concepts, from client interactions to project execution. Notably, the emphasis on Agile methodologies, client-centric approaches, and hands-on quality assurance practices provided

a comprehensive view of industry standards, contrasting the predominantly theoretical focus of our academic studies.

3. Personal Learning and Skill Development

Technical Skills Acquired:

- **Linux Proficiency:** Gained hands-on experience in utilizing Linux for software development tasks, enhancing command-line skills and system navigation.
- **Version Control:** Learned effective use of GitHub for collaborative coding, including branching, merging, and pull requests.
- **Quality Assurance:** Acquired practical knowledge in conducting various testing methodologies to ensure software reliability.

Soft Skills Developed

- **Professionalism:** Observed and adapted to corporate etiquette, including effective communication, punctuality, and teamwork.
- **Problem-Solving:** Enhanced analytical skills by tackling real-world challenges during assignments and projects.
- **Adaptability:** Learned to swiftly adjust to new tools, technologies, and workflows in a dynamic work environment.

4. Case Study: Impact of Industrial Attachment at Riseup Labs

Participating in the industrial training at Riseup Labs profoundly influenced my understanding of the software development landscape. Experiencing the full spectrum of the SDLC in a professional setting illuminated the intricacies of transforming client requirements into functional software solutions. Observing the company's collaborations with global entities like UNICEF, UNDP, and WHO underscored the significance of delivering high-quality, impactful software products. This attachment not only enriched my technical competencies but also instilled a deeper appreciation for the collaborative and iterative nature of software development in the industry.

About Riseup Labs:

Riseup Labs is a leading ISO 9001 & 27001 certified technology solutions provider based in Bangladesh. Since its inception in 2009, the company has specialized in delivering innovative and sustainable digital solutions, focusing on mobile and web application development. With a portfolio of over 700 projects, Riseup Labs has collaborated with esteemed organizations such as UNICEF, UNDP, USAID, FAO, WHO, and BBC, contributing significantly to global digital transformation initiatives.